

# Topic 5: Chemical energetics

## Premium Answer Booklet

|                |                      |                            |
|----------------|----------------------|----------------------------|
| 4<br>questions | 2020-2025<br>Paper 4 | Premium<br>readable layout |
|----------------|----------------------|----------------------------|

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

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## Question 01

5 marks

### 2020 May/June Paper 43 Question 6

Main topic: Topic 5: Chemical energetics

Cross-topic links: Topic 12: Experimental techniques and chemical analysis, Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry

Answer source label: 0620\_s20\_ms\_43.pdf

|                 |                                                                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6(a)(i)</b>  | <ul style="list-style-type: none"><li>• 2, 2, 4, 1</li></ul>                                                                                                                                                 |
| <b>6(a)(ii)</b> | <ul style="list-style-type: none"><li>• thermal decomposition</li></ul>                                                                                                                                      |
| <b>6(b)</b>     | <ul style="list-style-type: none"><li>• add aqueous sodium hydroxide</li><li>• then (reduction with) aluminium (foil) (and warm)</li><li>• (ammonia gas produced which) turns damp red litmus blue</li></ul> |

Easy explanation / reminder

- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates
- clearly stated.
- Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER

## Question 02

17 marks

### 2021 Oct/Nov Paper 42 Question 4

Main topic: Topic 5: Chemical energetics  
Answer source label: 0620\_w21\_ms\_42.pdf

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4(a)</b>     | <ul style="list-style-type: none"><li>• E of making bonds &gt; breaking bonds</li><li>• bond making releases energy</li><li>• bond breaking requires energy</li></ul>                                                                                                                                                                                                                                                                                                                                                                             |
| <b>4(b)(i)</b>  | <ul style="list-style-type: none"><li>• exothermic mark</li><li>• horizontal line below energy level to R.H.S. of reactants line</li><li>• and</li><li>• labelled <math>\text{COCl}_2(\text{g})</math> (1)</li><li>• activation E mark</li><li>• activation energy 'hump' with upward arrow labelled / activation energy (1)</li><li>• energy change mark</li><li>• one downward arrow labelled <math>\Delta H</math></li><li>• and</li><li>• energy change starting from E level of reactants and finishing at E level of products (1)</li></ul> |
| <b>4(b)(ii)</b> | <ul style="list-style-type: none"><li>• increases rate of reaction</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>4(c)(i)</b>  | <ul style="list-style-type: none"><li>• equilibrium shifts to right hand side (1)</li><li>• fewer moles (of gas) on right hand side (1)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>4(c)(ii)</b> | <ul style="list-style-type: none"><li>• equilibrium shifts to left hand side (1)</li><li>• (forward) reaction is exothermic (1)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>4(d)</b>     | <ul style="list-style-type: none"><li>• bond energy in making bonds</li><li>• <math>= [(2 \times 400) + 745] = 1545 \text{ (kJ mol}^{-1}\text{)}</math></li><li>• use of total E change</li><li>• <math>-230 = [240 + E(\text{C}\equiv\text{O})] - 1545</math></li><li>• OR</li><li>• <math>[240 + E(\text{C}\equiv\text{O})] = -230 + 1545 = (+1315)</math></li><li>• <math>E(\text{C}\equiv\text{O})</math></li><li>• <math>= [-230 + 1545] - 240 = 1075 \text{ (kJ mol}^{-1}\text{)}</math></li></ul>                                          |
| <b>4(e)</b>     | <ul style="list-style-type: none"><li>• all single bonding dot and cross pairs correct</li><li>• double <math>\text{C}=\text{O}</math> bond dot and cross pairs are correct</li><li>• complete diagram is correct</li></ul>                                                                                                                                                                                                                                                                                                                       |

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

## Question 03

16 marks

### 2023 Oct/Nov Paper 42 Question 5

Main topic: Topic 5: Chemical energetics  
Cross-topic links: Topic 6: Chemical reactions  
Answer source label: 0620\_w23\_ms\_42.pdf

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>5(a)</b>      | <ul style="list-style-type: none"><li>• enthalpy change (of reaction)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>5(b)(i)</b>   | <ul style="list-style-type: none"><li>• rate of reaction increases (1)</li><li>• more (CCl<sub>4</sub>) particles per unit volume (1)</li><li>• frequency of collisions between (CCl<sub>4</sub> and H<sub>2</sub>O) particles increases (1)</li></ul>                                                                                                                                                                                                                                                                                |
| <b>5(b)(ii)</b>  | <ul style="list-style-type: none"><li>• (position of) equilibrium moves to left hand side (1)</li><li>• more (gaseous) moles / molecules on right hand side (1)</li></ul>                                                                                                                                                                                                                                                                                                                                                             |
| <b>5(b)(iii)</b> | <ul style="list-style-type: none"><li>• Labels mark</li><li>• CCl<sub>4</sub>(g) + 2H<sub>2</sub>O(g) on reactant line</li><li>• AND</li><li>• CO<sub>2</sub>(g) + 4HCl(g) on product line</li><li>• Activation E mark</li><li>• upward arrow labelled E<sub>a</sub> from energy level of reactants to top of 'hump'</li><li>• Energy change mark</li><li>• one downward arrow labelled <math>\Delta H</math></li><li>• AND</li><li>• energy change starting from E level of reactants and finishing at E level of products</li></ul> |
| <b>5(b)(iv)</b>  | <ul style="list-style-type: none"><li>• minimum energy (1)</li><li>• that colliding particles must have to react (1)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>5(b)(v)</b>   | <ul style="list-style-type: none"><li>• (decreased by adding) catalyst</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>5(c)</b>      | <ul style="list-style-type: none"><li>• energy needed to break bonds in reactants</li><li>• [(4 <math>\rightarrow</math> 340) + (4 <math>\rightarrow</math> 460) = 3200 (kJ / mol) (1)</li><li>• energy released when bond in carbon dioxide form</li><li>• 2 <math>\rightarrow</math> 805 = 1610 (kJ / mol) (1)</li><li>• calculate H-Cl bond energy</li><li>• 3200 - (1610 + E(4H-Cl)) = - 130</li><li>• E(4H-Cl) = 3330 - 1610 = 1720 (kJ / mol) (1)</li><li>• 1720 / 4 = 430 (kJ / mol) (1)</li></ul>                             |

#### Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not
- escape.

END OF ANSWER

## Question 04

16 marks

### 2025 Oct/Nov Paper 41 Question 4

Main topic: Topic 5: Chemical energetics  
Answer source label: 0620\_w25\_ms\_41.pdf

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4(a)</b> | <ul style="list-style-type: none"><li>• energy level of products below energy level of reactants</li><li>• label of <math>\text{SO}_2(\text{g}) + 4\text{HF}(\text{g})</math> on product line</li><li>• upward arrow labelled <math>E_a</math> from energy level of reactants to the highest point of the curve</li><li>• one downward arrow labelled <math>\Delta H</math></li><li>• AND</li><li>• energy change starting from energy level of reactants and finishing at energy level of products</li></ul>                                                        |
| <b>4(b)</b> | <ul style="list-style-type: none"><li>• <math>= (4 \rightarrow \text{S-F}) + (4 \rightarrow \text{O-H}) = (4 \rightarrow 330) + (4 \rightarrow 460) = 3160</math></li><li>• bond energy of products - <math>3160 = -54</math></li><li>• <math>= 3160 - (-54) = 3214</math></li><li>• <math>(2 \rightarrow \text{S=O}) + (4 \rightarrow \text{H-F}) = 3214</math> therefore <math>2 \rightarrow \text{S=O} = 3214 - (4 \rightarrow 570) = 934</math></li><li>• <math>\text{S=O} = \frac{1}{2} \times 2 = 934</math> <math>\frac{1}{2} \times 2 = 467</math></li></ul> |
| <b>4(c)</b> | <ul style="list-style-type: none"><li>• increased temperature:</li><li>• (position of) equilibrium moves to left hand side</li><li>• reaction is exothermic</li><li>• increased pressure:</li><li>• (position of) equilibrium moves to left hand side</li><li>• less (gaseous) moles on left hand side</li><li>• no change / catalyst does not affect position of equilibrium</li></ul>                                                                                                                                                                              |
| <b>4(d)</b> | <ul style="list-style-type: none"><li>• kinetic energy of particles decreases</li><li>• frequency of collisions between particles decreases</li><li>• lower percentage/proportion/fraction of collisions / particles have energy greater than / equal to activation energy</li></ul>                                                                                                                                                                                                                                                                                 |

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.

END OF ANSWER